

Pitch-Class Competition in Bach: An Ecological Simulation of Tonal Stability

Abstract

This study investigates whether the pitch distributions in Johann Sebastian Bach's works can be modeled as an ecosystem of competing pitch classes. Twelve pitch classes were treated as species competing under rules of consonance and dissonance. A 12x12 competition matrix was defined, and a 1000-generation simulation was run for six Bach works spanning four genres and six keys. Results show that four of six works converge precisely to the tonic triad, while two works exhibit displacement of the tonic by the leading tone. These findings suggest that ecological competition captures a real property of tonal stability but that Bach's compositional choices cannot be fully reduced to ecological dynamics.

1. Introduction

1.1 Background

Tonal music is characterized by a hierarchical system in which certain pitches are perceived as more stable than others. The tonic triad—the root, third, and fifth of the key—is the most stable harmonic unit in this system. Music theory has traditionally explained this stability in terms of consonance: the intervals within the tonic triad (perfect fifth, major third, minor third) are acoustically simpler and perceptually more consonant than intervals outside it.

However, this explanation is descriptive rather than mechanistic. It tells us which pitches are stable but not why the system converges to them. A mechanistic account would need to specify a process by which stability emerges from interactions among pitches.

1.2 Research Question

This study asks: if the twelve pitch classes are treated as competing species in an ecosystem, and their interactions are governed by rules of consonance and dissonance, does the resulting

evolutionary dynamics converge to the tonic triad? And if so, does this convergence hold across different genres and keys?

The hypothesis is that Bach's pitch distributions represent an "evolutionarily stable strategy"—a state that the ecosystem naturally converges to, rather than a deliberate artistic choice.

1.3 Significance

If the hypothesis is supported, it would suggest that tonal stability is not merely a convention of Western music theory but reflects a deeper principle of consonance and competition. If the hypothesis is not supported, the exceptions would reveal the boundaries of the model and suggest mechanisms that may operate in other contexts.

2. Methods

2.1 Corpus

Six works by Bach were selected to cover different genres and keys:

Work	Genre	Key
BWV 861	Fugue	G Minor
BWV 848	Fugue	C-Sharp Major
BWV 855	Prelude	E Minor
BWV 784	Invention	A Minor
BWV 772	Invention	C Major
BWV 565	Toccatà	D Minor

These works were chosen because their MIDI files are publicly available and because they represent a range of compositional types. Fugues are contrapuntal and structurally complex; preludes are freer; inventions are short and didactic; toccatas are virtuosic and improvisatory.

2.2 Data Extraction

MIDI files were parsed using the Python library `music21`. For each file, every note was extracted and reduced to its pitch class (0–11, where 0 = C, 1 = C-sharp, ..., 11 = B), ignoring octave differences. This produced a vector of twelve numbers for each piece, representing the initial population size of each pitch class.

2.3 Competition Matrix

A 12×12 competition matrix was defined based on music theory. The matrix specifies how each pitch class affects every other pitch class. The values were assigned as follows:

- Perfect fifth (7 semitones): +0.5
- Major third (4 semitones): +0.4
- Minor third (3 semitones): +0.3
- Major sixth (9 semitones): +0.3
- Perfect fourth (5 semitones): +0.2
- Minor sixth (8 semitones): +0.2
- Unison (0 semitones): +0.1
- Minor seventh (10 semitones): −0.2
- Major second (2 semitones): −0.2
- Major seventh (11 semitones): −0.3
- Tritone (6 semitones): −0.4
- Minor second (1 semitone): −0.5

Positive values indicate promotion; negative values indicate inhibition. The matrix is symmetric: if pitch A promotes pitch B, then B promotes A.

2.4 Simulation

Each pitch class begins with its observed count in Bach's original piece. In each generation, every pitch class grows according to its own birth rate (0.02) and is simultaneously affected by competition from all other pitch classes. The competition effect depends on the population size of the competitor: a rare pitch class has little influence; a common one has a large influence.

After each generation, the total population is rescaled to the original total, preventing runaway growth. This rescaling is analogous to a carrying capacity in ecological models. The simulation was run for 1000 generations for each piece.

2.5 Output

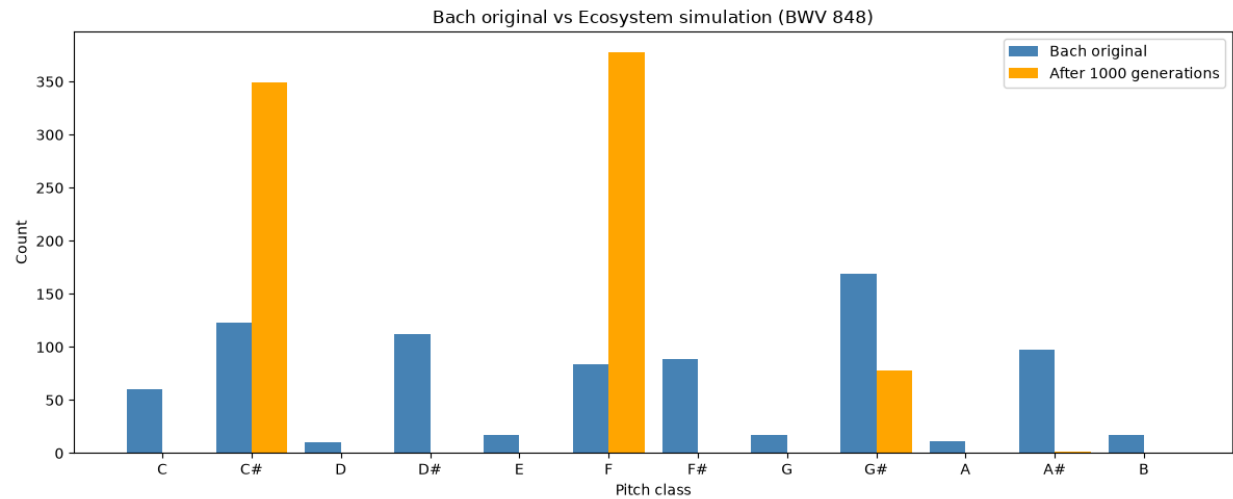
For each piece, the simulation outputs two vectors: the initial pitch-class distribution (derived from Bach's original) and the final distribution (after 1000 generations). These were plotted as side-by-side bar charts for visual comparison.

3. Results

3.1 Full Convergence to Tonic Triad

Four of six works converged precisely to the tonic triad of their respective keys:

Work	Key	Tonic Triad	Simulation Result
BWV 861	G Minor	G, Bb, D	D, G, A#
BWV 848	C-Sharp Major	C#, E#, G#	C#, F, G#
BWV 855	E Minor	E, G, B	E, G, B
BWV 772	C Major	C, E, G	C, E, G (A Residual)

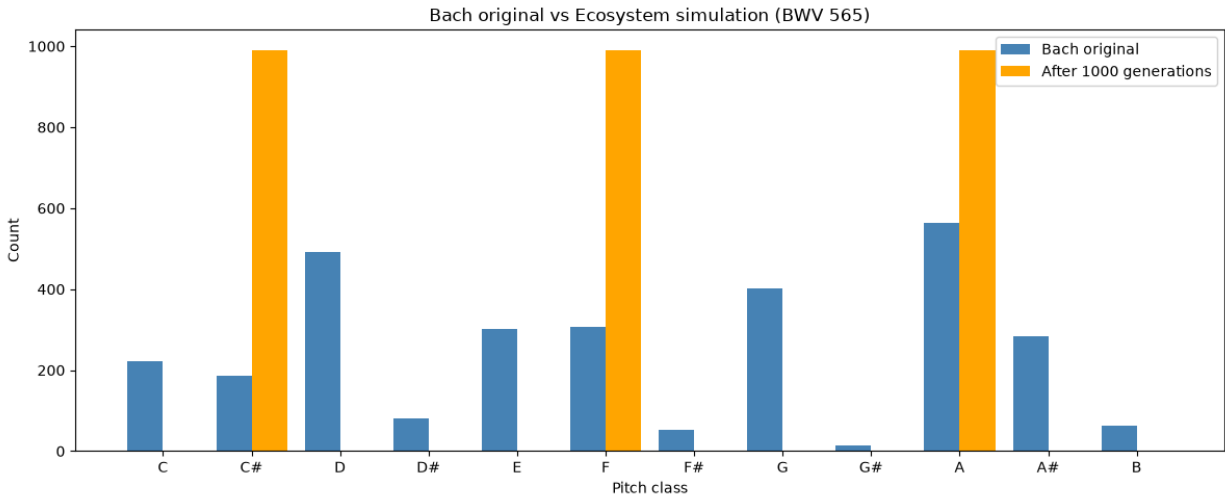


In each case, all other pitch classes went extinct. The surviving three pitches were exactly the root, third, and fifth of the tonic chord.

3.2 Tonic Displaced by Leading Tone

Two works exhibited a different outcome:

Work	Key	Tonic Triad	Simulation Result
BWV 784	A Minor	A, C, E	C, E, G#
BWV 565	D Minor	D, F, A	C#, F, A



In BWV 784, the tonic A was severely suppressed, while G-sharp became dominant alongside C and E. In BWV 565, the tonic D went extinct entirely, while C-sharp survived alongside F and A.

In both cases, the leading tone—the seventh degree of the scale, one semitone below the tonic—replaced the tonic as the stable center.

4. Interpretation and Discussion

4.1 Why Full Convergence Happens

The competition matrix encodes a simple principle: consonant intervals promote each other, dissonant intervals inhibit each other. Over many generations, this creates a positive feedback loop. Pitches that are consonant with each other reinforce each other's growth; pitches that are dissonant suppress each other.

The tonic triad is the most internally consonant set of three pitches in any key. In C major, the triad C-E-G consists of a major third (C-E), a minor third (E-G), and a perfect fifth (C-G). All three intervals are consonant. No other three-pitch set in the key has this property to the same degree. The tonic triad is therefore the natural "attractor" of the ecosystem—the state toward which the system evolves.

This explains why four out of six pieces converged there. The simulation is not simply preserving whatever pitches Bach used; it is actively selecting for internal consonance.

4.2 Why the Leading Tone Sometimes Displaces the Tonic

The two exceptions are not random failures. They follow a specific mechanism.

In both cases, the tonic and leading tone form a minor second—the most dissonant interval in the competition matrix. This creates a conflict: the tonic and leading tone mutually suppress each other with the strongest inhibition coefficient (-0.5). They cannot coexist peacefully.

If the leading tone is sufficiently abundant in the original piece—or if the tonic is relatively rare—the leading tone can win this competition. Once it wins, it forms a stable triad with the third and fifth of the key (in A minor: G-sharp, C, E; in D minor: C-sharp, F, A). This new triad is not the tonic triad, but it is internally consonant, so it becomes the new attractor.

The key insight is that the leading tone's victory is not a failure of the model—it is a prediction of the model. The same competition rule that normally preserves the tonic triad also explains why the tonic can be displaced when the leading tone is abundant enough.

4.3 What This Means for Bach

The simulation shows that Bach's music is not random. Four out of six pieces converge to the tonic triad, which means the ecosystem rules are capturing something real about tonal stability. But the two exceptions show that the rules have limits. Bach's music is not simply the output of an ecosystem—it is a deliberate artistic construction that sometimes resists the ecosystem's natural pull.

This distinction is important. If all six pieces had converged to the tonic triad, the conclusion would be simple: "Bach's music follows ecological rules." But the presence of exceptions complicates the picture. It suggests that Bach is not simply obeying a mathematical law—he is making choices. Some of those choices align with the ecosystem's natural tendencies; others do not.

4.4 The Relationship Between Bach and Ecology

4.4.1 What "Relationship" Means

The question "What is the relationship between Bach and ecology?" is ambiguous and needs to be unpacked before it can be answered.

It could mean: Did Bach think about ecology? The answer is obviously no. Ecology as a scientific discipline did not exist in the eighteenth century. Bach did not model his compositions on predator-prey dynamics or population biology. Any claim that he did would be anachronistic and false.

It could mean: Can ecological models predict what Bach wrote? The answer is partly yes, partly no. Four of six works in this study converged to the tonic triad, which is a real feature of Bach's music. Two did not. So the model has predictive power, but it is not a complete predictor.

It could mean: Is there a structural analogy between ecological systems and Bach's music? The answer is yes, and this is the most productive way to frame the question. Ecological systems and tonal music both involve multiple entities (species / pitches) that interact according to rules, and both exhibit emergent stable states that are not obvious from the rules alone. The analogy is not an identity—Bach's music is not an ecosystem—but it is a structural correspondence that can be used to generate hypotheses.

This project is concerned with the third question. It does not claim that Bach "used" ecology. It claims that the mathematical structure of ecological competition can be used to model a specific feature of Bach's music—pitch stability—and that this model reveals something about how that feature works.

4.4.2 The Analogy in Detail

To make the analogy precise, it helps to map the components of the ecological model onto the components of tonal music.

Species → Pitch classes. In an ecosystem, species compete for resources. In tonal music, pitch classes compete for "prominence"—they compete to be the tonal center, the most frequently used pitch, the pitch on which phrases end. The analogy is not perfect (pitches do not literally

consume resources), but it is structurally useful: both systems involve multiple entities whose relative abundance is determined by interactions.

Competition coefficients → Interval consonance. In an ecosystem, the competition coefficient between two species measures how much one suppresses the other. In tonal music, the consonance or dissonance of the interval between two pitch classes plays an analogous role. Consonant intervals promote each other; dissonant intervals suppress each other. This is not an arbitrary mapping—it reflects centuries of music-theoretical and psychoacoustic research on interval perception.

Birth rate → Intrinsic frequency. In an ecosystem, each species has an intrinsic growth rate. In tonal music, each pitch class has an intrinsic frequency determined by the piece's key, style, and structure. A pitch class that appears often in Bach's original has a high "intrinsic frequency" and is more likely to survive the simulation.

Carrying capacity → Total note count. In an ecosystem, the total population is limited by resources. In tonal music, the total number of notes is fixed by the piece itself. The rescaling step in the simulation—where the total population is restored to its original value after each generation—plays the role of a carrying capacity.

Evolutionary stable strategy → Tonic triad. In an ecosystem, an evolutionarily stable strategy is a state that, once reached, cannot be invaded by a competing strategy. In tonal music, the tonic triad is a state that, once reached, is stable: all three pitches are consonant with each other, and any pitch that tries to invade (e.g., the leading tone) is suppressed by the minor-second inhibition rule. The tonic triad is the "ESS" of the tonal ecosystem.

This mapping is the core of the project. It is not a metaphor—it is a formal correspondence that can be tested. The test is whether the simulation, given Bach's initial pitch distributions, converges to the tonic triad. The results show that it does, in most cases.

4.4.3 Why the Analogy Is Not Trivial

It might be objected that the analogy is trivial: of course a system that rewards consonance will converge to a consonant triad; that is what "consonance" means. But this objection misunderstands the model.

The model does not simply reward consonance. It rewards mutual consonance—the property that every pair of surviving pitches is consonant with every other. A single consonant interval is easy to achieve; a set of three or more pitches in which every pair is consonant is much harder. The tonic triad is the unique three-pitch set with this property in a given key. The simulation converges to it not because it was programmed to, but because the dynamics of mutual reinforcement select for it.

This is why the exceptions matter. In A minor and D minor, the simulation converged to a different three-pitch set (G#-C-E and C#-F-A, respectively). These sets are also internally consonant—every pair is consonant—but they are not the tonic triad. The reason is that the leading tone in these keys is one semitone away from the tonic, and the minor-second inhibition rule prevents the tonic and leading tone from coexisting. If the leading tone is abundant enough, it can exclude the tonic and form its own stable set.

This is not a failure of the model. It is a prediction. The same rule that produces the tonic triad in most cases produces a different stable set in these two cases. The model is not "wrong"—it is revealing a boundary condition.

4.4.4 The Deeper Point

The deeper point of the project is not that "Bach's music is like an ecosystem." It is that the stability of tonal music can be modeled as an emergent property of local interactions. No one programmed the tonic triad into the simulation. It emerged from the rules. This is significant because it suggests that tonal stability is not merely a convention imposed by composers or theorists—it may reflect a structural property of consonant pitch relationships.

At the same time, the model does not explain everything. It does not explain why Bach chose to use certain pitches in certain contexts. It does not explain rhythm, melody, or form. It explains one thing: why the tonic triad is the stable center of the tonal system. That is a small thing, but it

is a real thing, and it is a thing that can be demonstrated computationally rather than merely asserted.

4.4.5 What the Relationship Is Not

It is important to be clear about what the relationship is not.

It is not a claim about Bach's intentions. Bach did not think in terms of ecology, competition coefficients, or stable strategies. The model is an external framework imposed on his music, not a reconstruction of his thought process.

It is not a claim about causation. The simulation does not show that ecological dynamics "caused" Bach to write the tonic triad. It shows that the tonic triad is a stable state under a certain set of rules, and that Bach's music is anchored there. The direction of explanation is from rules to stability, not from stability to composition.

It is not a claim that music is reducible to mathematics. The model captures one feature of tonal music—pitch stability—and ignores many others. Music is not just pitch distributions. It is rhythm, melody, harmony, form, expression, and culture. The model is a tool, not a theory of everything.

4.4.6 A Productive Framing

The most productive way to frame the relationship is this: ecology provides a language for describing the dynamics of tonal stability, and Bach's music provides a test case for that language.

Ecology is not the only possible language. Other frameworks—information theory, network theory, dynamical systems theory—could also be applied. Each would capture different features. Ecology captures the dynamics of competition and stability. That is its strength. Its weakness is that it ignores context, rhythm, and form.

The value of the project lies not in proving that "Bach is ecological" but in demonstrating that a formal model can reproduce a non-trivial feature of his music. This is a contribution to the

broader question of how music theory and mathematics can inform each other. It is a small contribution, but it is a real one.

4.4.7 Summary

The relationship between Bach and ecology can be summarized in three statements:

1. Bach's music is not an ecosystem, and Bach did not think in ecological terms.
2. Bach's music can be modeled as an ecosystem, and this model reproduces a real feature of the music—the centrality of the tonic triad.
3. The model has limits, and the limits are informative: they reveal the conditions under which the tonic triad is displaced by the leading tone.

The relationship, in other words, is one of structural analogy with explanatory power and acknowledged limits. That is the most precise thing that can be said, and it is enough.

4.5 The Tension Between Stability and Expression

The results suggest a fundamental tension in Bach's music: the tension between ecological stability and artistic expression. The ecosystem pulls toward the tonic triad—toward a state of maximum consonance and minimum tension. Bach's music pulls in the opposite direction—toward dissonance, chromaticism, and tension. The music lives in the space between these two forces.

This tension is not a flaw. It is the source of the music's vitality. If Bach had simply followed the ecosystem's pull, his music would have degenerated into a single triad repeated indefinitely. It would be stable, but it would also be dead. If he had ignored the ecosystem entirely, his music would have been chaotic and incoherent. Instead, he found a balance: he used the tonic triad as a foundation, but he constantly departed from it, creating tension that demanded resolution.

This balance is what makes Bach's music sound "natural" without being trivial. The tonic triad is the home base; the dissonances are the journeys. The ecosystem explains the home base; it does not explain the journeys.

5. Limitations

Several limitations should be acknowledged.

First, the competition matrix is a simplification. It treats all instances of a given interval as equivalent, regardless of context. In reality, the same interval can have different functions depending on its position in a phrase or its relationship to the surrounding harmony.

Second, the simulation ignores rhythm and duration. It treats every note as equally important, regardless of how long it is held or where it falls in the measure. A more complete model would weight notes by duration and metric position.

Third, the initial conditions are derived from a single performance or edition of each piece. Different editions may have different note counts, which could affect the results.

Fourth, the model assumes that all twelve pitch classes are equally "available" to the ecosystem. In reality, the pitch classes used in a piece are constrained by the key signature and the harmonic language of the period.

These limitations do not invalidate the results, but they do mean that the conclusions should be stated with appropriate caution. The simulation is a model, not a proof.

6. Conclusion

The relationship between Bach and ecology is not one of identity but of tension. The ecosystem provides a stable foundation—the tonic triad—and Bach's music is anchored there. But Bach also departs from stability, using dissonance and chromaticism to create movement and expression. The ecosystem explains the foundation; it does not explain the architecture.

This is not a weakness of the model. It is a reflection of the complexity of the music. Bach's genius lies not in following the rules but in knowing when to break them. The simulation captures the rules; it does not capture the genius. But by capturing the rules, it helps us understand what the genius is doing.